

Serin Hong

Personal Information

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Citizenship: South Korea (U.S. permanent resident)

Academic Positions

University of Arizona

Assistant Professor, August 2023 – present

Simons Laufer Mathematical Sciences Institute (SLMath/MSRI)

Simons Bridge Postdoctoral Fellow, August 2022 – July 2023

University of Michigan

Research Fellow, August 2022 – July 2023

Postdoctoral Assistant Professor, September 2018 – July 2022

Education

California Institute of Technology

Ph.D. in Mathematics, June 2018

- Advisor: Elena Mantovan
- Mandatory military service for South Korea, July 2010 – July 2013

M.S. in Mathematics, June 2010 (awarded June 2011)

Stanford University

M.S. in Electrical Engineering, June 2009

B.S. in Mathematics with Honors, June 2008

Awards and Honors

2025 – 2030	Simons Travel Support
2022 – 2023	Simons Bridge Postdoctoral Fellowship
2022	Early-career AMS-NSF-Simons-ICM Travel Grant (canceled due to the Russo-Ukrainian war)
2021	Juha Heinonen Award for Excellence in Postdoctoral Teaching
2019 – 2021	Honored Instructor
2019	Oberwolfach Leibniz Graduate Students Grant
2018	Scott Russell Johnson Graduate Dissertation Prize
2016	Apostol Award for Excellence in Teaching
2014	Scott Russell Johnson Prize for Excellence in Graduate Research
2010	Scott Russell Johnson Prize for Excellence in First-Year Graduate Studies
2008	J. E. Wallace Sterling Award for Scholastic Achievement <i>for the top 25 graduating seniors in the School of Humanities and Sciences</i>
2008	Highbridge Award for Mathematical Problem Solving
2005 – 2008	William Lowell Putnam Mathematical Competition <i>Top 25 individuals (2005, 2008), Top 5 teams (2007, 2008)</i>
2004 – 2009	Samsung Scholarship
2003	International Mathematical Olympiad <i>Silver Medal</i>

Research Articles

- [12] *Filtrations of symplectic vector bundles on the Fargues-Fontaine curve*, preprint
- [11] *Constructing vector-valued automorphic forms for unitary groups* (with T. Browning, P. Coupek, E. Eischen, C. Frechette, S. Y. Lee, and D. Marcil), Doc. Math., to appear
- [10] *On nonemptiness of Newton strata in the B_{dR}^+ -Grassmannian for $GS_{p,n}$* , preprint.
- [9] *On nonemptiness of Newton strata in the B_{dR}^+ -Grassmannian for GL_n* , Doc. Math. (2024)
- [8] *Extensions of vector bundles on the Fargues-Fontaine curve II*, J. Algebraic Geom. (2025)
- [7] *On certain extensions of vector bundles in p -adic geometry*, Math. Res. Lett. (2023)
- [6] *Classification of subbundles on the Fargues-Fontaine curve*, Algebra & Number Theory. (2021)
- [5] *Classification of quotient bundles on the Fargues-Fontaine curve*, Selecta Math. (2023)
- [4] *Extensions of vector bundles on the Fargues-Fontaine curve* (with C. Birkbeck, T. Feng, D. Hansen, Q. Li, A. Wang, and L. Ye), J. Inst. Math. Jussieu. (2022).
- [3] *Harris-Viehmann conjecture for Hodge-Newton reducible Rapoport-Zink spaces*, J. London Math. Soc. (2018).
- [2] *On the Hodge-Newton filtration for p -divisible groups of Hodge type*, Math. Z. (2019).
- [1] *On Hodge-Newton reducible local Shimura data of Hodge type*, Caltech Ph.D. dissertation (2018).

Books

- [1] *Lecture Notes on p -adic Hodge Theory*, World Scientific, in press.

Conference Talks/Lecture Series

- 2026 *PMI Intensive Lecture Series*, Postech, Korea
- 2024 *Number Theory Intensive Lecture Series*, Postech, Korea
- 2024 *Southern California Number Theory Day*, UC San Diego, CA
- 2021 *KMS Annual Meeting*, online
- 2021 *SNU Special Lecture Series*, Seoul National University, Korea
- 2021 *PMI Intensive Lecture Series*, Postech, Korea
- 2020 *AMS Sectional Meeting*, Purdue University, IN (canceled due to COVID-19 outbreak)
- 2019 *Upstate Number Theory Conference*, Cornell University, NY (contributed)
- 2018 *Number Theory/Topology Mini Workshop*, KAIST, Korea

Invited Seminar Talks

- 2026 *Korea University*, Algebra Special Seminar
- 2026 *Yonsei University*, Number Theory Seminar
- 2024 *Colorado State*, FRAGMENT Seminar
- 2024 *Postech*, Number Theory Seminar
- 2023 *Univ. of Arizona*, Algebra and Number Theory Seminar
- 2023 *Univ. of Maryland*, Lie Group and Representation Theory seminar
- 2022 *Caltech*, Number Theory Seminar
- 2022 *Seoul National University*, Number Theory Seminar
- 2022 *Postech*, Number Theory Seminar
- 2022 *Purdue*, Automorphic Form Seminar
- 2021 *Univ. of Arizona*, Algebra and Number Theory Seminar
- 2021 *Postech*, Number Theory Seminar
- 2020 *KIAS*, Number Theory Seminar
- 2020 *Univ. of Michigan*, Group, Lie and Number Theory Seminar
- 2019 *Seoul National University*, Number Theory Seminar
- 2019 *Binghamton*, The Arithmetic Seminar
- 2018 *Univ. of Michigan*, Group, Lie and Number Theory Seminar

2017 *Caltech*, Number Theory Seminar
 2017 *UCLA*, Number Theory Seminar
 2017 *UC San Diego*, Number Theory Seminar
 2016 *Caltech*, Number Theory Seminar

Graduate Students Advised

2030 exp. Hrishikesh Ghatе, Ph.D.
 2028 exp. Napoleon Wang, Ph.D.
 2028 exp. Ameya Borwankar, Ph.D.
 2025 Pavel Chumarov, M.S.
 2024 Haris Serdarevic, M.S.
 2021 Maxim Melnik, M.S. (joint with Urs Hartl)

Teaching Experience

University of Arizona

Instructor

Math 511: Algebra	Fall 2025 – Spring 2026
Math 223: Vector Calculus	Spring 2025
Math 519: Introduction to p -adic Hodge theory	Fall 2024
Math 313: Introduction to Linear Algebra	Fall 2023 – Spring 2024

University of Michigan

Instructor

Math 215: Multi-variable Calculus	Fall 2020 – Winter 2022
Math 679: Introduction to p -adic Hodge theory	Winter 2020
Math 115: Single-variable Calculus	Fall 2018 – Winter 2020

California Institute of Technology

Instructor

Math 17: Putnam Problem Solving Seminar	Fall 2015, Fall 2016
Math 7: Number Theory for Beginners	Spring 2016

Teaching Assistant

ACM 95: Applied Mathematics for the Physical Sciences	Winter 2018
ACM/EE 117: Probability and Random Process	Fall 2017
Math 3: Probability and Statistics	Winter 2014, Winter 2016
Math 7: Number Theory for Beginners	Spring 2015
Math 120: Graduate Algebra	Fall 2014 – Winter 2015
Math 1: Calculus of One and Several Variables	Fall 2009 – Spring 2010

Community Services and Outreach Programs

2022	University of Michigan African Presidential Scholars (UMAPS)
2020 - 2022	Ann Arbor Community Resources (AACR)
2019	Postdoctoral Mentorship for graduate students

Department Services

2025 - present	Computer committee
2024 - present	Co-organizer for Algebra and Number Theory Seminar
2024 - present	Graduate committee
2023 - 2024	Colloquium Chair
2015 - 2016	Head coach for the Putnam competition team

Additional Activities

Co-organizer for the Arizona Winter School (2024 - present)

Referee for peer reviewed journals (e.g., Compositio, Duke, Nagoya Math, PAMS, Res. Num. Th)

Reviewer for UKRI Grants and Simons Travel Support

Reviewer for Mathematical Reviews / MathSciNet

Co-organizer for the Arithmetic Geometry Learning Seminar at Caltech (2017)

Tutor/Mentor for a summer camp for mathematically gifted Korean middle school students (2013)

Author of the textbook “Creative Thinking and Mathematics” for Korean middle schools (2013)

Developer of an education program for mathematically gifted Korean middle school students (2012)

Lecturer at the Korean Mathematical Olympiad Summer/Winter School (2004-2005)